

Section I

10 marks

Attempt Question 1 to 10

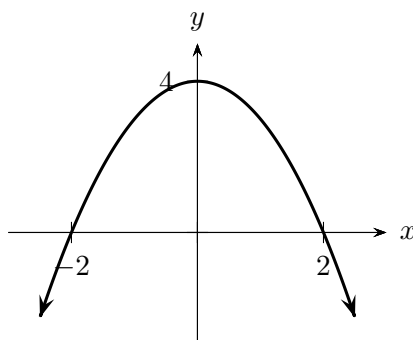
Allow approximately 10 minutes for this section

Mark your answers on the answer sheet provided.

Questions

Marks

1. Which equation corresponds to the graph below? 1



- (A) $y = 2 - x^2$ (B) $y = 4 - x^2$ (C) $y = x + 4$ (D) $y = x^2 + 4$

2. The point P divides the interval $A\left(\frac{17}{3}, 2\right)$ and $B(-3, 4)$ externally in the ratio $2 : 3$. Which of the following are the coordinates of P ? 1

- (A) $(-23, 2)$ (B) $(-9, -12)$ (C) $(9, 0)$ (D) $(23, -2)$

3. Which of the following conditions for $\frac{dP}{dt}$ and $\frac{d^2P}{dt^2}$ describe the slowing growth of a variable P ? 1

(A) $\frac{dP}{dt} > 0$ and $\frac{d^2P}{dt} > 0$. (C) $\frac{dP}{dt} > 0$ and $\frac{d^2P}{dt} < 0$.

(B) $\frac{dP}{dt} < 0$ and $\frac{d^2P}{dt} < 0$. (D) $\frac{dP}{dt} < 0$ and $\frac{d^2P}{dt} > 0$.

4. Which of the following best describes the function $f(x) = \frac{x^2 - 1}{x}$? 1

- (A) Even function (C) Neither odd nor even function
(B) Odd function (D) Zero function

5. Which of the following is an expression for $\frac{d}{dx} \left(x\sqrt{x^2+2} \right)$? 1

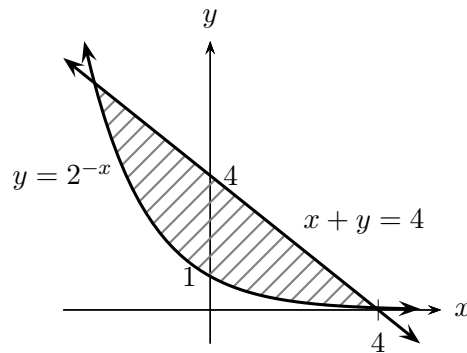
(A) $\frac{x}{\sqrt{x^2+2}}$

(C) $\frac{2x^2+2}{\sqrt{x^2+2}}$

(B) $1 + \frac{x}{x^2+2}$

(D) $\frac{2x^2+x+4}{\sqrt{x^2+2}}$

6. Which pair of inequalities defines the shaded region? 1



(A) $\begin{cases} x+y \geq 4 \\ y \geq 2^{-x} \end{cases}$

(C) $\begin{cases} x+y \geq 4 \\ y \leq 2^{-x} \end{cases}$

(B) $\begin{cases} x+y \leq 4 \\ y \geq 2^{-x} \end{cases}$

(D) $\begin{cases} x+y \leq 4 \\ y \leq 2^{-x} \end{cases}$

7. If $3^a = 7^b$, what is the value of $9^{\frac{a}{b}}$? 1

(A) 0.56

(B) 1.77

(C) 3.46

(D) 49

8. Which of the following exists at the point $(1,1)$ on the graph of $y = (x-1)^3 + 1$? 1

(A) A local minimum.

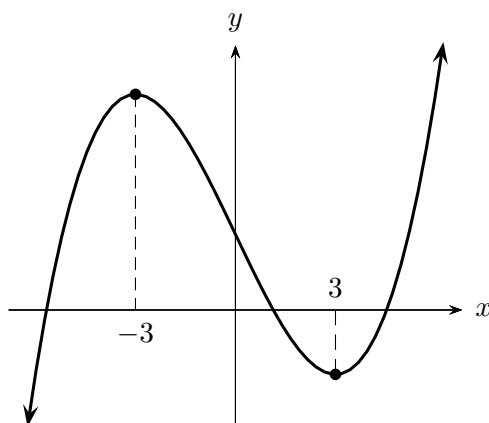
(C) A stationary point of inflexion.

(B) A local maximum.

(D) None of the above.

9. From the graph of $y = f(x)$, when is $f'(x)$ negative?

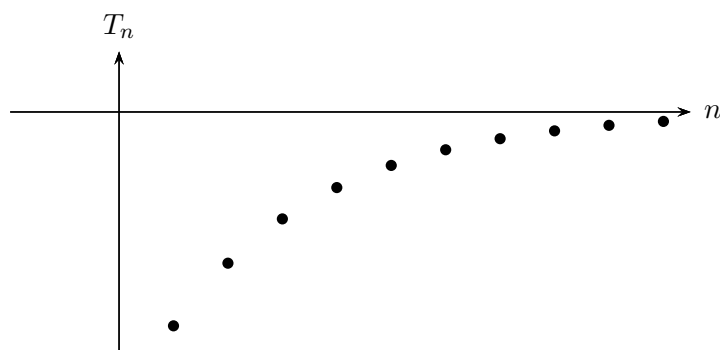
1



- (A) $x < -3$ or $x > 3$ (C) $x \leq -3$ or $x \geq 3$
 (B) $-3 < x < 3$ (D) $-3 \leq x \leq 3$

10. The graph shows consecutive terms of a sequence. Which of the following statements best describes the sequence?

1



- (A) geometric, $|r| \geq 1$. (C) arithmetic, $|d| \geq 1$.
 (B) geometric, $|r| < 1$. (D) arithmetic, $|d| < 1$.

Examination continues overleaf...

Section II

90 marks

Attempt Questions 11 to 16

Allow approximately 1 hour and 50 minutes for this section.

Write your answers in the writing booklets supplied. Additional writing booklets are available. Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 Marks)	Commence a NEW page.	Marks
(a) Fully factorise: $x^2 - 2x + 1 - 4y^2$		2
(b) Fully simplify: $\sqrt{\frac{3^{5k+2}}{27^k}}$		2
(c) Solve $x^2 + 2x - 8 > 0$.		2
(d) Solve $ x + 3 < 2$.		2
(e) Evaluate $\lim_{x \rightarrow -1} \frac{x^2 - 2x - 3}{2x^2 - x - 3}$.		2
(f) Fully simplify: $\frac{\log(a^2b^2) - \log(ab^2)}{\log \sqrt{a}}$.		2
(g) Solve the following inequation: $\frac{-1}{x(x-2)} < 1$.		3

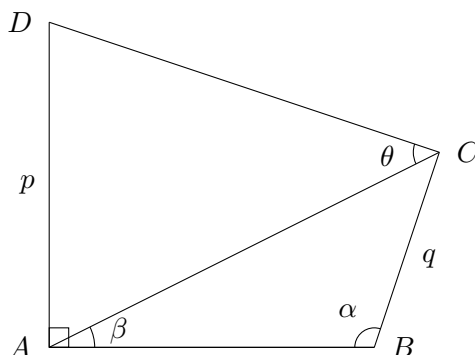
Question 12 (15 Marks)

Commence a NEW page.

Marks

- (a) Solve $\tan^2 \theta = \frac{1}{3}$ for $0^\circ \leq \theta \leq 360^\circ$. **3**
- (b) If $\tan \theta = -\frac{4}{5}$ and $\cos \theta > 0$, find the exact value of $\sin \theta$. **2**
- (c) Prove that $\frac{\sin \theta}{1 - \cos \theta} = \frac{1 + \cos \theta}{\sin \theta}$. **2**
- (d) $ABCD$ is a quadrilateral with AD perpendicular to AB .

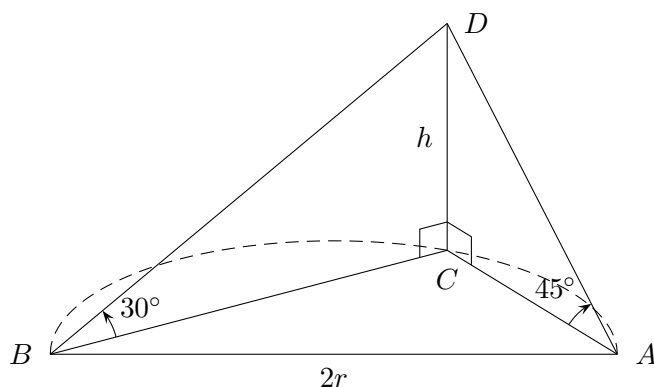
Given that $\angle CAB = \beta$, $\angle ABC = \alpha$, $\angle ACD = \theta$, $AD = p$ and $BC = q$,



- i. Show that $\angle ADC = 90^\circ - (\theta - \beta)$. **2**
- ii. Using the sine rule, prove that **3**

$$q = \frac{p \sin \beta \cos(\theta - \beta)}{\sin \theta \sin \alpha}$$

- (e) AB is the diameter of a semicircular block of land with radius r metres. CD is a vertical flagpole of height h metres standing on its base with C on the arc AB . From A and B , the angles of elevation to the top of the flagpole (D) are 45° and 30° respectively. **3**



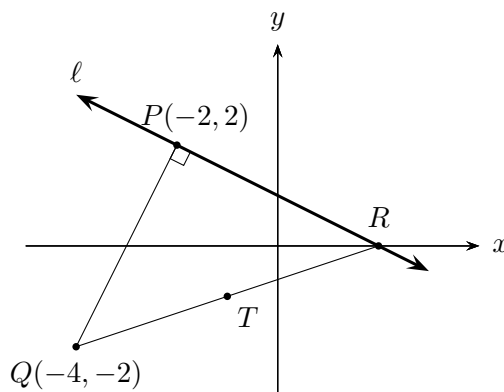
Show that $h = r$.

Question 13 (15 Marks)

Commence a NEW page.

Marks

- (a) In the diagram, $P(-2, 2)$ and $Q(-4, -2)$ are marked.



- i. Show that the equation of the line ℓ through P , perpendicular to PQ is given by $x + 2y - 2 = 0$ 1
 - ii. The line ℓ intersects the x axis at R . Find the coordinates of R . 1
 - iii. Show that the midpoint T , of QR is $(-1, -1)$. 1
 - iv. Find the perpendicular distance from T to the interval PR . 2
- (b) Consider the series $721, 708, 695 \dots$
- i. Show that this forms an arithmetic series. 1
 - ii. Show that $T_n = 734 - 13n$, where T_n is the n -th term of the series. 2
 - iii. Find the value of the first negative term in the series. 2
- (c) Given x and $y > 0$,
- $x, -2$ and y forms a geometric series
 - $-2, y$ and x forms an arithmetic series.
- i. Find the value of xy . 1
 - ii. Find the value(s) of x and y . 2
 - iii. Find the limiting sum (where it exists) of the geometric series 2

$$x, -2, y \dots$$

Question 14 (15 Marks)

Commence a NEW page.

Marks

- (a) Differentiate the following functions with respect to
- x
- .

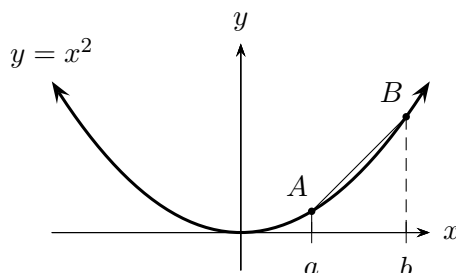
i. $y = x^5 + 5x^2 - 6x + 7.$ **1**

ii. $y = (3x - 1)^5.$ **1**

iii. $y = \frac{x - 1}{x^2 + 1}.$ **2**

iv. $y = \frac{1}{2\sqrt{x}}.$ **2**

- (b) In the diagram,
- A
- and
- B
- are points on the parabola
- $y = x^2$
- . The
- x
- coordinate of
- A
- is
- a
- , and the
- x
- coordinate of
- B
- is
- b
- .

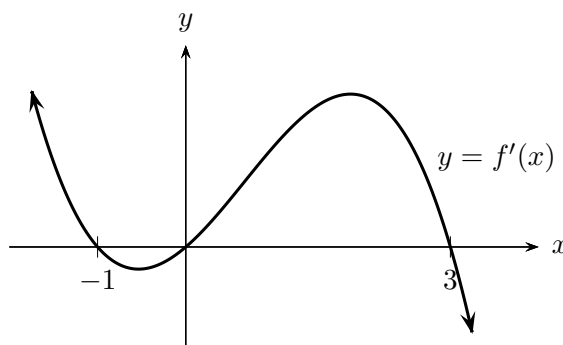


- i. Find the gradient of the chord
- AB
- in terms of
- a
- and
- b
- .
- 1**

- ii. What is the limiting value of the gradient of the chord
- AB
- as
- b
- approaches
- a
- ?
- 1**

- iii. What is the geometric significance of this limiting value?
- 1**

- (c) The diagram shows the graph of
- $y = f'(x)$
- for a certain function
- $y = f(x)$
- .
- 3**



If $f(x)$ has exactly two zeros, sketch a possible graph of $y = f(x)$. Do *not* find the values of the intercepts or the y values of the stationary points.

- (d) The normal to the curve
- $y = \sqrt{x}$
- at the point
- P
- on the curve has an angle of inclination of
- 135°
- with the positive
- x
- axis.
- 3**

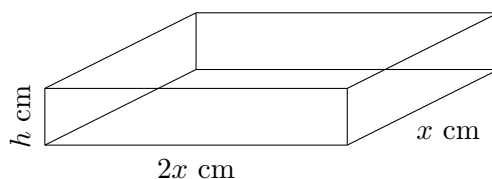
Find the coordinates of the point P .

Question 15 (15 Marks)

Commence a NEW page.

Marks

- (a) Consider the function $f(x) = x^4 - 4x^3 + 1$.
- Find the coordinates of the stationary points of $f(x)$, and determine their nature. **4**
 - Find the coordinates of the points of inflexion. **2**
 - Draw a neat sketch of the curve $y = f(x)$, showing the above features. **2**
- (b) Joe is building a small *open topped* toy box, indicated by the diagram below. The box is twice as long as it is wide. The box has a total external surface area of $3\,750\text{ cm}^2$. **Note:** the box does not have a lid.



- Show that the height of the box is given by **2**
- $$h = \frac{625}{x} - \frac{x}{3}$$
- Find the dimensions of the box which would give the maximum volume. **3**
 - Joe decides the height of the box cannot exceed $10\frac{5}{6}\text{ cm}$. **2**

Find the new dimensions of the box if the surface area was to remain at $3\,750\text{ cm}^2$.

Question 16 (15 Marks)

Commence a NEW page.

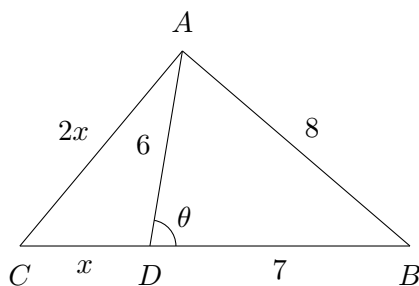
Marks

- (a) The position x metres at time t seconds of a particle moving in a straight line is given by

$$x = 2t^3 - 9t^2 + 12t + 6$$

- i. Find the times when the velocity is zero. **2**
- ii. Find the average velocity between $t = 0$ and $t = 1$. **1**

- (b) The diagram shows $\triangle ABC$ with D on CB such that $AD = 6$, $DB = 7$, $AB = 8$, $CD = x$, $AC = 2x$ and $\angle ADB = \theta$.

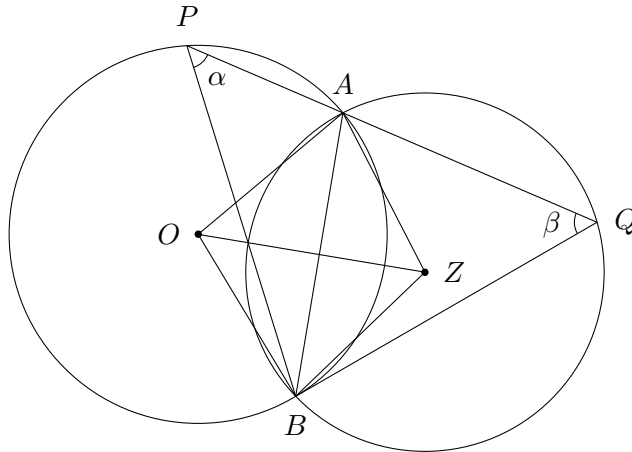


- i. Show that $\cos \theta = \frac{1}{4}$. **1**
- ii. Hence or otherwise, find the value of x . **3**

Question 16 continued on the next page...

Question 16 continued from the previous page...

- (c) A line through A meets the two circles (with centres O and Z) again at P and Q . Let $\angle BPQ = \alpha$ and $\angle AQB = \beta$.



Copy or trace the diagram into your writing booklet.

Hint: all parts can be completed without additional constructions.

- | | |
|--|----------|
| i. Prove that $\triangle AOZ \equiv \triangle BOZ$. | 3 |
| ii. Prove that OZ bisects $\angle AOB$ and $\angle AZB$. | 2 |
| iii. Hence prove that $\angle BOZ = \alpha$ and $\angle BZO = \beta$. | 2 |
| iv. Prove that $\angle PBQ = \angle OBZ$. | 1 |

End of paper.

BLANK PAGE

Answer sheet for Section I

Mark answers to Section I by fully blackening the correct circle, e.g “●”

STUDENT NUMBER:

Class (please ✓)

☐ 11M3A – Mr Ireland

☐ 11M3D – Mr Lam

☐ 11M3B – Mr Berry

☐ 11M3E – Mr Lowe

☐ 11M3C – Ms Ziazaris

☐ 11M3F – Mr Choy

- 1 – ☐ A ☐ B ☐ C ☐ D
- 2 – ☐ A ☐ B ☐ C ☐ D
- 3 – ☐ A ☐ B ☐ C ☐ D
- 4 – ☐ A ☐ B ☐ C ☐ D
- 5 – ☐ A ☐ B ☐ C ☐ D
- 6 – ☐ A ☐ B ☐ C ☐ D
- 7 – ☐ A ☐ B ☐ C ☐ D
- 8 – ☐ A ☐ B ☐ C ☐ D
- 9 – ☐ A ☐ B ☐ C ☐ D
- 10 – ☐ A ☐ B ☐ C ☐ D

Suggested Solutions

(f) (2 marks)

Section I

1. (B) 2. (D) 3. (C) 4. (B) 5. (C)
 6. (B) 7. (D) 8. (C) 9. (B) 10. (B)

Question 11 (Choy)

(a) (2 marks)

- ✓ [1] for significant progress towards answer.
 ✓ [1] for final answer.

$$\begin{aligned} & x^2 - 2x + 1 - 4y^2 \\ &= (x-1)^2 - 4y^2 \\ &= (x-1-2y)(x-1+2y) \end{aligned}$$

(b) (2 marks)

- ✓ [1] for significant progress towards answer.
 ✓ [1] for final answer.

$$\begin{aligned} \sqrt{\frac{3^{5k+2}}{27^k}} &= \sqrt{\frac{3^{5k+2}}{3^{3k}}} \\ &= \left(3^{2k+2}\right)^{\frac{1}{2}} \\ &= 3^{k+1} \end{aligned}$$

(c) (2 marks)

$$\begin{aligned} & x^2 + 2x - 8 > 0 \\ & (x+4)(x-2) > 0 \\ & x < -4 \text{ or } x > 2 \end{aligned}$$

(d) (2 marks)

$$\begin{aligned} & |x+3| < 2 \\ & -2 < x+3 < 2 \\ & -5 < x < -1 \end{aligned}$$

(e) (2 marks)

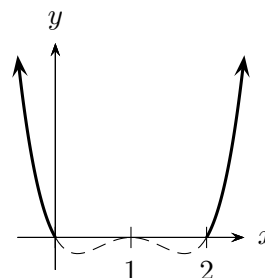
$$\begin{aligned} & \lim_{x \rightarrow -1} \frac{x^2 - 2x - 3}{2x^2 - x - 3} \\ &= \lim_{x \rightarrow -1} \frac{(x-3)\cancel{(x+1)}}{(2x-3)\cancel{(x+1)}} \\ &= \frac{-4}{-5} = \frac{4}{5} \end{aligned}$$

$$\begin{aligned} & \frac{\log(a^2b^2) - \log(ab^2)}{\log\sqrt{a}} \\ &= \frac{\log\left(\frac{a^2b^2}{b^2}\right)}{\frac{1}{2}\log a} \\ &= \frac{2\log a}{\log a} = 2 \end{aligned}$$

(g) (3 marks)

- ✓ [1] for eliminating denominator with unknown.
 ✓ [1] for further significant progress towards answer.
 ✓ [1] for final answer.

$$\begin{aligned} & \frac{-1}{x(x-2)} < \frac{1}{x^2(x-2)^2} \\ & -x(x-2) < x^2(x-2)^2 \\ & x^2(x-2)^2 + x(x-2) > 0 \\ & x(x-2)(x(x-2)+1) > 0 \\ & x(x-2)(x^2-2x+1) > 0 \\ & x(x-2)(x-1)^2 > 0 \end{aligned}$$



$$\therefore x < 0 \text{ or } x > 2$$

Question 12 (Ziaziaris)

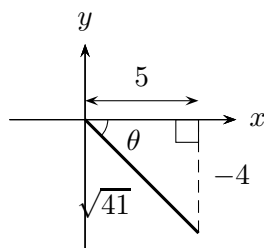
(a) (3 marks)

- ✓ [1] for significant progress towards answer.
- ✓ [1] for each pair of answers.

$$\begin{aligned}\tan^2 \theta &= \frac{1}{3} \\ \tan \theta &= \pm \frac{1}{\sqrt{3}} \\ \therefore \theta &= 30^\circ, 150^\circ, 210^\circ, 330^\circ\end{aligned}$$

(b) (2 marks)

$$\tan \theta = -\frac{4}{5} \quad \cos \theta > 0$$



$$\therefore \sin \theta = -\frac{4}{\sqrt{41}}$$

(c) (2 marks)

$$\begin{aligned}\frac{\sin \theta}{1 - \cos \theta} \times \frac{(1 + \cos \theta)}{(1 + \cos \theta)} &= \frac{\sin \theta(1 + \cos \theta)}{1 - \cos^2 \theta} \\ &= \frac{\sin \theta(1 + \cos \theta)}{\sin^2 \theta} \\ &= \frac{1 + \cos \theta}{\sin \theta}\end{aligned}$$

(d) i. (2 marks)

- ✓ [1] for significant progress towards answer.
- ✓ [1] for final answer.
- $\angle DAC = 90^\circ - \beta$
(complementary $\angle$)
- $\therefore \angle ADC = 180^\circ - \theta - (90^\circ - \beta)$
($\angle$ sum of $\triangle ADC$).
 $= 90^\circ - (\theta - \beta)$.

ii. (3 marks)

- ✓ [1] for correct application of sine rule
- ✓ [1] for further significant progress towards answer.
- ✓ [1] for final answer.
- In $\triangle ABC$,

$$\begin{aligned}\frac{q}{\sin \beta} &= \frac{AC}{\sin \alpha} \\ AC &= \frac{q \sin \alpha}{\sin \beta}\end{aligned}\quad (1)$$

- In $\triangle ADC$,

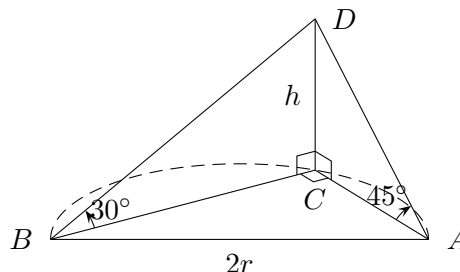
$$\begin{aligned}\frac{p}{\sin \theta} &= \frac{AC}{\sin(90^\circ - (\theta - \beta))} \\ \frac{p \cos(\theta - \beta)}{\sin \theta} &= AC\end{aligned}\quad (2)$$

- Equate (1) and (2)

$$\begin{aligned}\frac{q \sin \alpha}{\sin \beta} &= \frac{p \cos(\theta - \beta)}{\sin \theta} \\ \therefore q &= \frac{p \sin \beta \cos(\theta - \beta)}{\sin \theta \sin \alpha}\end{aligned}$$

(e) (3 marks)

- ✓ [1] for each correct dot point.



- $\angle BCA = 90^\circ$ ($\angle$ in a semicircle)
- In $\triangle BCD$,

$$\begin{aligned}\tan 30^\circ &= \frac{h}{BC} \\ BC &= \frac{h}{\tan 30^\circ} = h\sqrt{3}\end{aligned}$$

Similarly,

$$AC = \frac{h}{\tan 45^\circ} = h$$

- In $\triangle ABC$,

$$\begin{aligned}AC^2 + BC^2 &= AB^2 \\ (2r)^2 &= h^2 + 3h^2 \\ 4h^2 &= 4r^2 \\ \therefore r &= h\end{aligned}$$

Question 13 (Berry)

(a) i. (1 mark)

✓ [1] All working must be shown to obtain full mark.

$$m_{PQ} = \frac{2 - (-2)}{-2 - (-4)} = \frac{4}{2} = 2$$

Equation of line ℓ : $m_{\perp} = -\frac{1}{2}$

$$y = -\frac{1}{2}x + b$$

As ℓ passes through $(-2, 2)$,

$$2 = -\frac{1}{2}(-2) + b$$

$$b = 1$$

$$\therefore y = -\frac{1}{2}x + 1$$

$$-2y = x - 2$$

$$x + 2y - 2 = 0$$

ii. (1 mark) When $y = 0$,

$$x + 2(0) - 2 = 0$$

$$\therefore x = 2$$

$$\therefore R(2, 0)$$

iii. (1 mark)

$$R(2, 0) \quad Q(-4, -2)$$

$$\begin{aligned} T &= \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right) \\ &= \left(\frac{2 + (-4)}{2}, \frac{0 + (-2)}{2} \right) = (-1, -1) \end{aligned}$$

iv. (2 marks)

$$PR: x + 2y - 2 = 0 \quad T(-1, -1)$$

$$\begin{aligned} d_{\perp} &= \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}} \\ &= \frac{|1(-1) + 2(-1) - 2|}{\sqrt{1^2 + 2^2}} \\ &= \frac{|-1 - 2 - 2|}{\sqrt{1 + 4}} \\ &= \frac{5}{\sqrt{5}} = \sqrt{5} \end{aligned}$$

(b) i. (1 mark) If series is arithmetic,

$$T_2 - T_1 = 708 - 721 = -13$$

$$T_3 - T_2 = 695 - 708 = -13$$

Hence series is arithmetic.

ii. (2 marks)

$$T_n = 721 - 13(n - 1)$$

$$= 721 - 13n + 13 = 734 - 13n$$

iii. (2 marks)

The first negative term occurs when $T_n < 0$.

$$\therefore 734 - 13n < 0$$

$$\frac{13n}{\div 13} > \frac{734}{\div 13}$$

$$n > \frac{734}{13} = 56.46 \dots$$

$$\therefore n = 57$$

$$T_{57} = 734 - 13(57) = -7$$

(c) i. (1 mark) For a GP, $\frac{T_2}{T_1} = \frac{T_3}{T_2}$.

$$\frac{-2}{x} = \frac{y}{-2}$$

$$\therefore xy = 4$$

ii. (2 marks)

For the AP,

$$T_2 - T_1 = T_3 - T_2$$

$$y - (-2) = x - y$$

$$2y = x - 2$$

As $xy = 4$, then $y = \frac{4}{x}$

$$2\left(\frac{4}{x}\right) = x - 2$$

$$8 = x^2 - 2x$$

$$x^2 - 2x - 8 = 0$$

$$(x - 4)(x + 2) = 0$$

$$\therefore x = 4, -2$$

$$y = 1, -2$$

Since x and $y > 0$, then $x = 4$ and $y = 1$ only.

iii. (2 marks)

Series: 4, -2, 1, with $r = -\frac{1}{2} < 1$.

$$S = \frac{a}{1 - r} = \frac{4}{1 - (-\frac{1}{2})} = \frac{8}{3}$$

Question 14 (Ireland)

(a) i. (1 mark)

$$\frac{dy}{dx} = 5x^4 + 10x - 6$$

ii. (1 mark)

$$\frac{dy}{dx} = 5(3x - 1)^4 \times 3 = 15(3x - 1)^4$$

iii. (2 marks)

$$y = \frac{x-1}{x^2+1}$$

$$u = x-1 \quad v = x^2+1$$

$$u' = 1 \quad v' = 2x$$

$$\frac{dy}{dx} = \frac{(x^2+1) - 2x(x-1)}{(x^2+1)^2}$$

iv. (2 marks)

$$y = \frac{1}{2}x^{-\frac{1}{2}}$$

$$\frac{dy}{dx} = -\frac{1}{4}x^{-\frac{3}{2}}$$

(b) i. (1 mark)

$$m_{AB} = \frac{A(a, a^2) - B(b, b^2)}{b - a} = \frac{(b-a)(b+a)}{b-a}$$

$$= a + b$$

ii. (1 mark)

As $b \rightarrow a$, $m_{AB} \rightarrow 2a$.

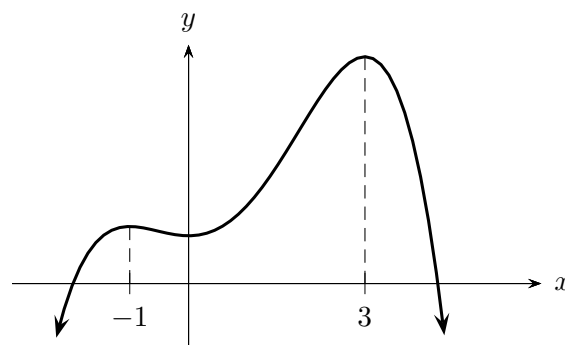
iii. (1 mark)

Value of the limit which leads to gradient of tangent to $y = x^2$ at $x = a$.

(c) (3 marks)

- ✓ [3] local max/min in correct locations, two zeros indicated, correct x values, no 'kinks', graph neat and clear.
- ✓ [2] quartic, and one of the criterion above is absent.
- ✓ [1] shows $y' = 0$ (max/min).

x	-1	0	3
$\frac{dy}{dx}$	+	0	-
y	↗	↘	↗

Example:

(d) (3 marks)

- ✓ [1] for correct differentiation.
- ✓ [1] for $m_{\perp}$.
- ✓ [1] for point.

$$y = x^{\frac{1}{2}}$$

$$m_{\tan} = \frac{dy}{dx} = \frac{1}{2}x^{-\frac{1}{2}}$$

As angle of inclination is 135° ,

$$m_{\perp} = \tan 135^\circ = -1$$

$$\therefore m_{\tan} = 1$$

$$\therefore \frac{1}{2}x^{-\frac{1}{2}} = 1$$

$$\frac{1}{2\sqrt{x}} = 1$$

$$2\sqrt{x} = 1$$

$$\sqrt{x} = \frac{1}{2}$$

$$x = \frac{1}{4} \text{ (note domain: } x \geq 0)$$

$$\therefore y = \sqrt{\frac{1}{4}} = \frac{1}{2}$$

$$\therefore P\left(\frac{1}{4}, \frac{1}{2}\right)$$

Question 15 (Lowe)

iii. (2 marks)

(a) i. (4 marks)

✓ [1] for each correct coordinate.

✓ [1] for each correct classification.

$$f(x) = x^4 - 4x^3 + 1$$

$$f'(x) = 4x^3 - 12x^2 = 4x^2(x - 3)$$

$$f''(x) = 12x^2 - 24x = 12x(x - 2)$$

Stationary points occur when $f'(x) = 0$:

$$f'(x) = 0$$

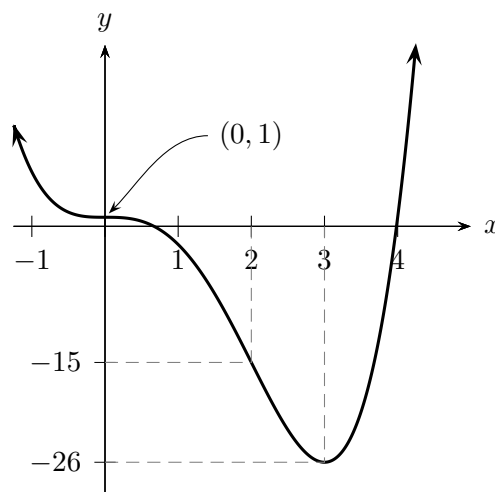
$$4x^2(x - 3) = 0$$

$$x = 0, 3$$

$$y = 1, -26$$

x	-1	0	1	3	4
$\frac{dy}{dx}$	-	0	-	0	+
y	↘ 1 ↘ -26 ↗				

Hence $(-1, 1)$ is a horizontal point of inflexion, and $(3, -26)$ is a local minimum. (Alternatively, check stationary points via second derivative)



ii. (2 marks)

Points of inflexion occur when $f''(x) = 0$:

$$12x(x - 2) = 0$$

$$\therefore x = 0, 2$$

x	-1	0	1	2	3
$\frac{d^2y}{dx^2}$	+	0	-	0	+
	∪		∩		∪

As there are changes in concavity at $x = 0$ and $x = 2$, therefore $(0, 1)$ is a horizontal point of inflexion and $(2, -15)$ is a point of inflexion.

(b) i. (2 marks)

$$SA = 3\,750 = 2x^2 + 2(2xh) + 2(xh)$$

$$2x^2 + 6xh = 3\,750$$

$$x^2 + 3xh = 1\,875$$

$$3xh = 1\,875 - x^2$$

$$h = \frac{1\,875 - x^2}{3x} = \frac{625}{x} - \frac{x}{3}$$

ii. (3 marks)

- ✓ [1] for finding $x = 25$.
- ✓ [1] for checking for local max.
- ✓ [1] for dimensions.

$$\begin{aligned} V &= 2x^2h \\ &= 2x^2 \left(\frac{625}{x} - \frac{x}{3} \right) \\ &= 1\,250x - \frac{2}{3}x^3 \end{aligned}$$

Differentiating w.r.t x ,

$$\frac{dV}{dx} = 1\,250 - 2x^2$$

Stationary points occur when $\frac{dV}{dx} = 0$:

$$\begin{aligned} 2x^2 &= 1\,250 \\ x^2 &= 625 \quad \Rightarrow \quad x = 25 \end{aligned}$$

Test for local maximum at $x = 25$:

$$\frac{d^2V}{dx^2} = -4x$$

When $x > 0$, $\frac{d^2V}{dx^2} < 0$. Hence $x = 25$ is a local maximum. Finding dimensions:

$$h = \frac{625}{25} - \frac{25}{3} = \frac{50}{3}$$

Hence dimensions are 50 cm, $\frac{50}{3}$ cm, 25 cm

iii. (2 marks)

$$\text{If } h < 10\frac{5}{6} \left(= \frac{65}{6} \right),$$

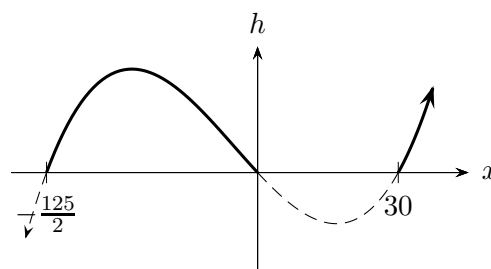
$$\underbrace{\frac{625}{x} - \frac{x}{3}}_{\times 6x^2} < \frac{65}{6}$$

$$3\,750x - 2x^3 < 65x^2$$

$$2x^3 + 65x^2 - 3\,750x > 0$$

$$x(2x^2 + 65x - 3\,750) > 0$$

$$x(2x + 125)(x - 30) > 0$$



From the inequality, for $h < \frac{65}{6}$, $x > 30$ (and boundary condition of $x > 0$ applies as x is a length). Hence $x = 30$, $2x = 60$ and $h = \frac{65}{6}$.

Check:

$$\frac{625}{30} - \frac{30}{3} = \frac{125}{6} - 10 = 10\frac{5}{6}$$

Question 16 (Lam)

(a) i. (2 marks)

$$x = 2t^3 - 9t^2 + 12t + 6$$

$$\frac{dx}{dt} = 6t^2 - 18t + 12$$

$$= 6(t^2 - 3t + 2)$$

$$= 6(t-2)(t-1)$$

$$\therefore \frac{dx}{dt} = 0 \text{ when } t = 1, 2$$

ii. (1 mark)

$$v_{\text{avg}} = \frac{x_2 - x_1}{t_2 - t_1}$$

$$x(0) = 6$$

$$x(1) = 2 - 9 + 12 + 6 = 11$$

$$\therefore v_{\text{avg}} = \frac{11 - 6}{1 - 0} = 5 \text{ ms}^{-1}$$

(b) i. (1 mark)

In $\triangle ADB$,

$$\begin{aligned} \cos \theta &= \frac{6^2 + 7^2 - 8^2}{2 \times 6 \times 7} \\ &= \frac{21}{84} = \frac{1}{4} \end{aligned}$$

ii. (3 marks)

✓ [1] for $\cos(180^\circ - \theta) = -\cos \theta$

✓ [1] for correct application of cosine rule.

✓ [1] for final answer.

In $\triangle ADC$,

$$\begin{aligned} \cos(180^\circ - \theta) &= \frac{x^2 + 6^2 - (2x)^2}{2 \times x \times 6} \\ &= \frac{-3x^2 + 36}{12x} \\ &= \frac{12 - x^2}{4x} \end{aligned}$$

As $\cos(180^\circ - \theta) = -\cos \theta$,

$$-\frac{1}{4} = \frac{12 - x^2}{4x}$$

$$-x = 12 - x^2$$

$$x^2 - x - 12 = 0$$

$$(x+3)(x-4) = 0$$

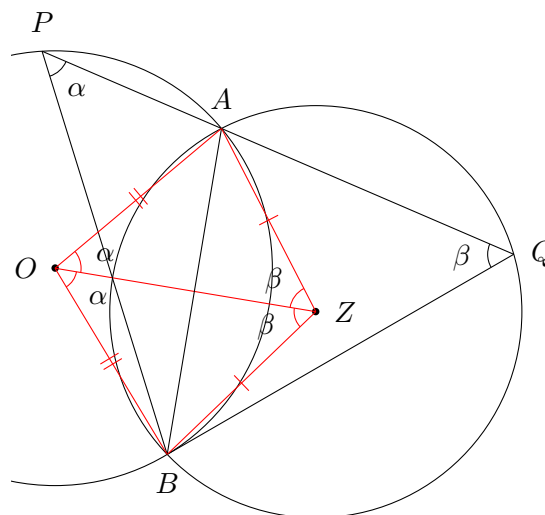
$$\therefore x = 4$$

(as x is a length)

(c) i. (3 marks)

✓ [2] for body of proof.

✓ [1] for final answer.

In $\triangle AOZ$ and $\triangle BOZ$,

- $AO = BO$ (equal radii)
- $AZ = BZ$ (equal radii)
- OZ is common

 $\therefore \triangle AOZ \equiv \triangle BOZ$ (SSS)

ii. (2 marks) As

$$\angle AOZ = \angle BOZ$$

(corresponding $\angle$ in congruent triangles), and similarly

$$\angle AZO = \angle BZO$$

then OZ must bisect $\angle AOB$ and $\angle AZB$.

iii. (2 marks)

- $\angle AOB = 2\alpha$ ($\angle$ at the centre is twice the $\angle$ at the circumference, subtended by the same arc)
- Since OZ bisects $\angle AOB$, then $\angle BOZ = \alpha$; this applies to $\angle BZO$ but with β as the angle.

iv. (1 mark)

- In $\triangle PBQ$, $\angle PBQ = 180^\circ - \alpha - \beta$ ($\angle$ sum of $\triangle$)
- In $\triangle OBZ$, $\angle OBZ = 180^\circ - \alpha - \beta$ ($\angle$ sum of $\triangle$)
- Hence $\angle PBQ = \angle OBZ$.